

# ST2304 - Statistical Modelling for Biologists/Biotechnologists

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# This week we will...

## Start the course

- ▶ admin
- ▶ try to motivate you
- ▶ overview of the course

## Start learning R

- ▶ introduction
- ▶ hands-on work

# Administration Matters

(we will deal with these in more detail later)

- ▶ Reference Group
- ▶ Blackboard
- ▶ web page: <https://wiki.math.ntnu.no/st2304/2026v/start>
- ▶ Material: <https://www.math.ntnu.no/emner/ST2304/2026v/>
- ▶ GitHub: <https://oharar.github.io/ST2304/>

# How the Course Will Run: Modules

One module a week

Modules on web pages

- ▶ text
- ▶ exercises (with hints)
- ▶ short recorded lectures

Active Learning, Group work, Problem solving

# How the Course Will Run: Contact time

Any announcements at the start

In effect everything will be exercise sessions in groups

- ▶ “Lectures” mainly for the modules
- ▶ Exercise session will be mainly for the exercises

Work in groups, ask for help when you want it.

# Assessment

Complete 8 exercise sets (of about 10)

- ▶ do in groups
- ▶ pass/fail
- ▶ first couple of weeks won't count
  - ▶ we will tell you when they start to count

# Resources

## Blackboard

- ▶ announcements
- ▶ links to more material
- ▶ exercises

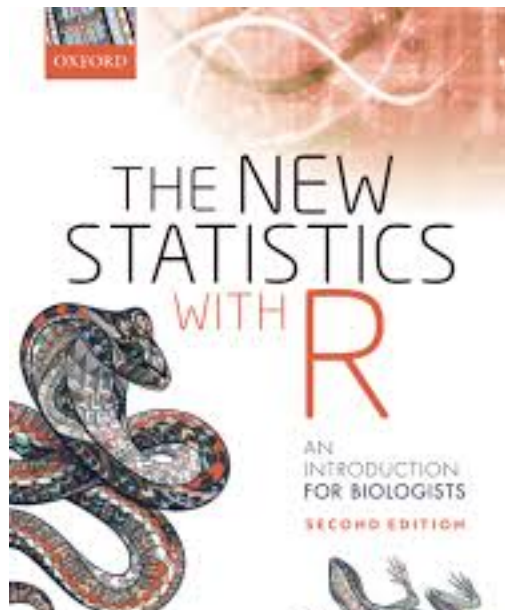
Web page: <https://wiki.math.ntnu.no/st2304/2026v/start> (this includes other links from this presentation)

- ▶ modules

## Text books

## Text Books

New Statistics with R - Andy Hector





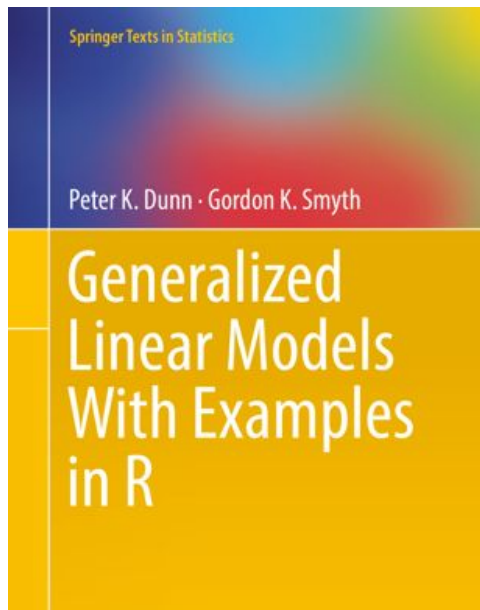
## Text Books

The Analysis of Biological Data - Whitlock & Schluter



## Text Books

Generalized Linear Models With Examples in R - Dunn & Smyth



## Other Resources

Blackboard

Wiki

Google (yes, use it!)

## Other Resources: AI

Like Google, AI can be a tool to help you learn

If you use it just to get answers for the exercises, you probably won't learn anything.

For the exercises, if you use AI explain how you use it

## Recap: What do you remember from ST0103?

Discuss in groups. Come up with 3 topics you learned about, and for each give an example where they are used in biology

- ▶ Add the topics to TaskCard:



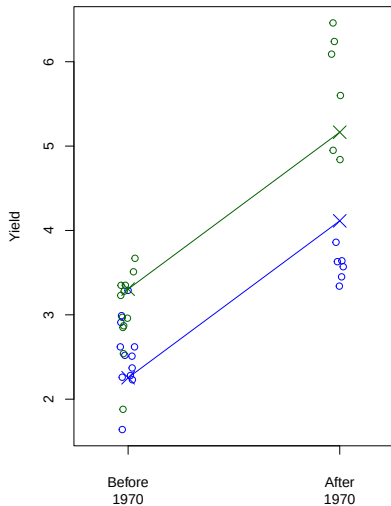
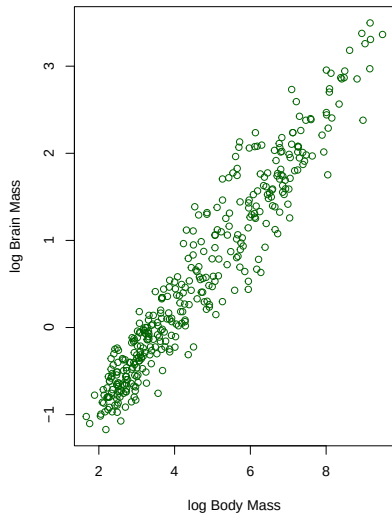


# What we are aiming for

We want you to be able to analyse your own data (and understand what you are doing!)

- ▶ fit the right models to data
- ▶ assess if the model is any good
- ▶ compare models and decide which is 'best'
- ▶ interpret the models

# Types of model I: Linear models (regression, ANOVA)

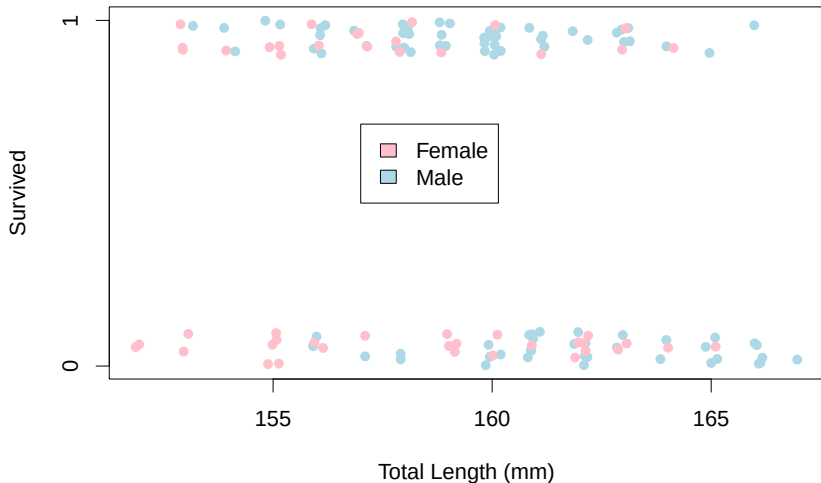




# Types of model I: Generalised Linear models

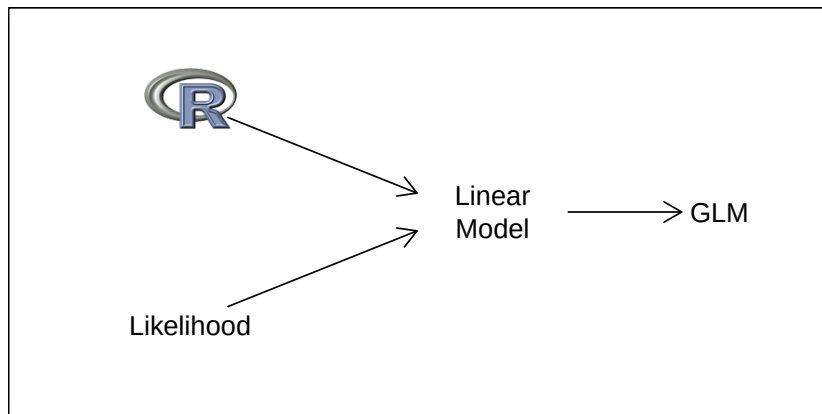
When things aren't normal

- ▶ binary (e.g. survive/died)
- ▶ counts (e.g. how many sparrows are there?)



## How do we get there?

Need some theory (likelihood) and to know how to write the models in the computer (R)



Then can start modelling. linear models is complicated regression

GLMs are complicated linear models

# Likelihood

The statistical framework to do this

Likelihood = probability of the data

- ▶ means we can write everything as probabilities

# R

The stats package we will use

- ▶ free, most commonly used
- ▶ more shortly

# Course Structure

Weeks 1-3: Likelihood and R

- ▶ statistical theory, and programming
- ▶ the background you will need to understand what follows, and to do it

Weeks 4-10: Linear models (regression, ANOVA)

- ▶ fitting straight lines

Weeks 11-13: Generalised Linear models

- ▶ fitting straight lines to different types of data